Introduction to Digital Systems
K. Sridharan

GREETINGS FOR THE NEW YEAR !

Time and Venue

- Class Time
  - B-slot: Mon (9 am), Tue (8 am),
    Wed (1 pm) and Fri (11 am)
    Normally, 1 tutorial per week
    (mostly the Friday slot)
- Venue
  - ESB 106

What is a Digital System ?

- Question for Students

What are some examples of digital systems ?

- Question for students

Digital Systems

- There is no single way to precisely describe/characterize a digital system.
- One way is to think of it as a piece of hardware that processes signals that have a finite number of discrete values.
- Another is to think of it as a system that comprises of hardware and software to process inputs and store the results suitably.

Examples

- Computers (desktop, laptop)
- Smartphones
- Digital thermometer

By the way …

- Digital does not necessarily mean binary !
- Often, however, we deal with quantities that have just two values (namely, 0 and 1).
- Digital computers use binary numbers.
- The initial representations can, in general, be in other forms (decimal, octal etc.)

What else exists ?

- Question for students

What else exists ?

- Besides digital systems, we also have analog systems.
- Analog systems process signals that can take any value in principle across a continuous range of voltage, current or some other quantity.
- Early radios, televisions and telephones were analog.

More on digital systems

- A digital system like a computer has many components and broadly, they are of two types: combinational logic elements and sequential logic elements.
- In this course, we will discuss different number systems, algebra that is appropriate and various elements that go into a digital system in detail.

Overview of the course

- Introduction to number systems and Boolean algebra
- Logic minimization and implementation
- Combinational logic
- Sequential logic
- State machine design
- Asynchronous circuits

Textbook and References

- M. Mano and M. Ciletti, Digital Design, Pearson Education, 2018
- C. Roth, Fundamentals of Logic Design, Thomson Brooks/Cole, 2005
- S. Brown and Z. Vranesic, Fundamentals of digital logic with VHDL design, McGraw Hill, 2017.

Assessment

- 2 quizzes as per schedule of the academic section, each for 20 marks
- Tutorial submissions (10 marks)
- End-semester examination (also as per schedule of the academic section) for 50 marks

Moodle

- Please ensure that you can access Moodle (https://courses.iitm.ac.in)
- Announcements will be made via Moodle (and in class – so do NOT miss class!)

Concerns/Feedback…

- Can ask questions during/after lectures
- We welcome feedback !